Logistic Regression:

→ Linear reg predicts the numerical response but is not suitable for predicting the categorical variables.

→ When categorical variables are involved, it is called classification problem

→ L.R is suitable for binary classification problem.

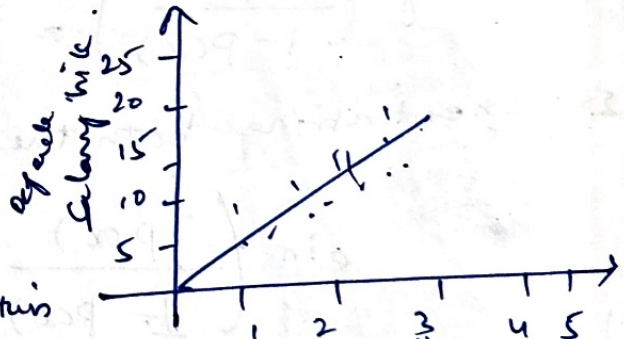
Eg = Is the mail spam or not spam? Yes | No

(Binary responses)

- Logistic Regression Algorithm work?

→ whether an employ would get a promotion or not based on their performance?

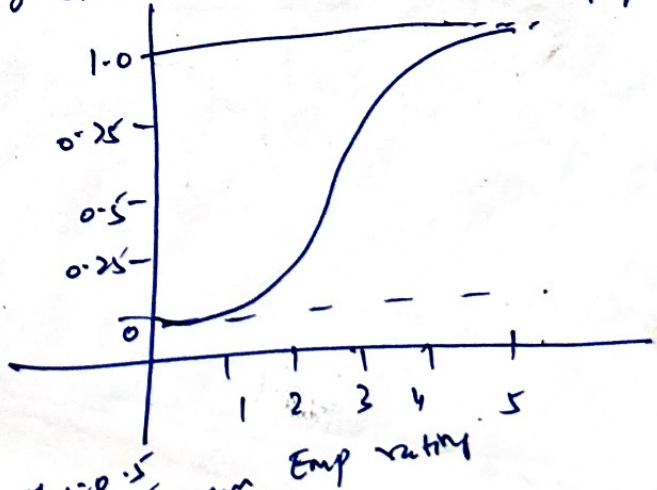
→ Linear graph won't be suitable in this case.



(Linear Regression)

- we clip the line at zero and one, Convert into a sigmoid curve (S-curve)

→ Based on threshold value, org decide whether employee will get salary hike or not.



To understand logistic regression, Let's go over the odds.

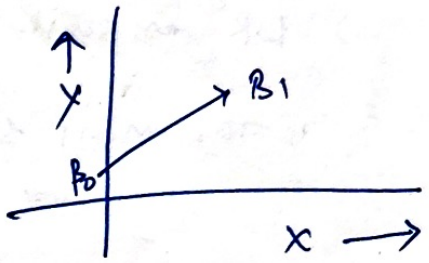
- $\text{odds}(\Omega) = \dfrac{\text{probability of an event happening (Success) } (p)}{\text{probability of an event not happening } (1-p)}$

- $\text{odds}(\Omega) = \dfrac{p}{1-p}$

- The value of odds range from zero to ∞ and the value of prob lies Btw 0 to 1

LHS = RHS

$y = \beta_0 + \beta_1 x$



- Now to predicts the odds of success.

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$$

β_0 - Intercept

β_1 - Coefficient of Regression.

- Exponentiating both the sides:

$$\text{ein}\left(\frac{p(x)}{1-p(x)}\right) = e^{\beta_0+\beta_1 x}$$

$$\left(\frac{p(x)}{1-p(x)}\right) = e^{\beta_0+\beta_1 x}$$

$$e^{\ln(x)} = x$$

$y = e^{B_0 + B_1 * x}$

then $\frac{p(x)}{1-p(x)} = y$

$p(x) = y(1-p(x))$

$p(x) = y - y(p(x))$

$p(x) + y(p(x)) = y$

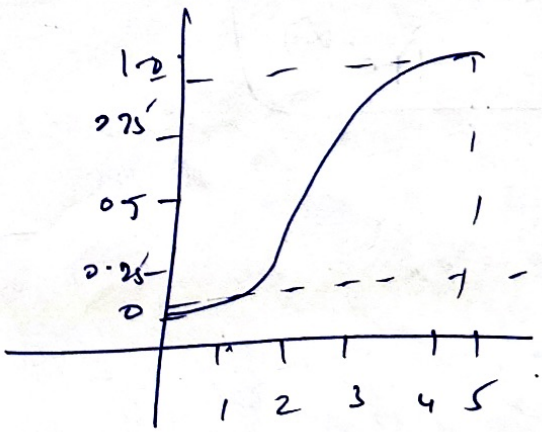
$p(x)(1+y) = y$

$p(x) = \frac{y}{1+y}$

$p(x) = \frac{e^{B_0+B_1 x}}{1+e^{B_0+B_1 x}}$ $\left\{ \cdot \frac{e^{-(B_0+B_1 x)}}{e^{-(B_0+B_1 x)}} \right.$

$= \frac{1}{1+e^{-(B_0+B_1 x)}}$

$p(x) = \frac{1}{1+e^{-(B_0+B_1 x)}}$



Eg: The std dataset has Interview marks Based on history of those who are selected or not)

- Based on the LR, the value of the Learnt parameters are $\beta_0 = 1$ and $\beta_1 = 8$, Assuming ~~Marks~~ $X = 60$, Compute the resultant class.

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

$$= \beta_0 + \beta_1 x = 481$$

$$= \frac{1}{1 + e^{-481}} = 0.44$$

$$\therefore th = 0.5$$

$$= 0.44 < 0.5$$

$$\boxed{S(x) = \frac{1}{1 + e^{-x}}}$$

Logistic Regression:

Problem Statement.

→ The Dataset of pass or fail in Exam of 5 Students in given in the table.

- use LR as classifier to Answer.

Hours Study	Pass(1) / Fail(0)
29	0
15	0
33	1
28	1
39	1

① Calculate the probability of Pass for the Student who studies 33 hours.

② At least How many hours student should study that makes he will Pass the courses with probability of more than 95-1.

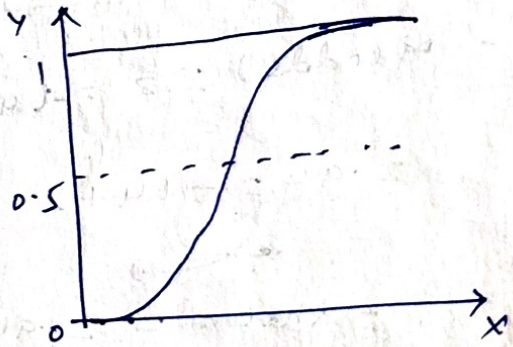
∴ Assumed model suggested by the optimizer for odds of passing the course is

$$\log(\text{odds}) = -64 + 2 * \text{hours}$$

↑ Independent Variable.

- use Sigmoid function

$$S(x) = \frac{1}{1+e^{-x}}$$



Sol:

q ① - probability of pass for the student who studies 33 hours.

$$p = \frac{1}{1+e^{-z}} \qquad S(x) = \frac{1}{1+e^{-x}}$$

$$\log(\text{odds}) = z = -64 + 2 * \text{hours}$$

$$z = -64 + 2 \times 33 = -64 + 66 = 2 \checkmark$$

$$p = \frac{1}{1+e^{-2}} = 0.88$$

- That's, if student studies 33 hours, there is 88% Chance that the student will Pass the Exam

Q2 At least how many hours student should that makes the will Pass the Courese with the probability of more than 95-1.

$$P = \frac{1}{1+e^{-z}} = 0.95 \quad (95\text{-}1)$$

Natural ordering

$\therefore \ln(e^x) = x$

$$= 0.95 * (1 + e^{-z}) = 1$$

$$0.95 * e^{-z} = 1 - 0.95 \quad (\leftarrow 0.95 * 0.95 e^{-2} = 1)$$

$$e^{-z} = \frac{0.05}{0.95} = 0.0526$$

$$\ln(e^{-z}) = \ln(0.0526)$$

$$-z = \ln(0.0526) = -2.94$$

$$\boxed{z = 2.94}$$

$$\log(\text{odds}) = z = -64 + 2 * \text{hours}$$

$$2.94 = -64 + 2 * \text{hours}$$

$$2 * \text{hours} = 2.94 + 64$$

$$2 * \text{hours} = 66.94$$

$$\text{hours} = \frac{66.94}{2}$$

$$\text{hours} = 33.47 \text{ hours}$$

Probability:

The stud should study at least 33.47 hours, so that he pass the examination 95-1. more than pass means in probability